MATH840 Time Series

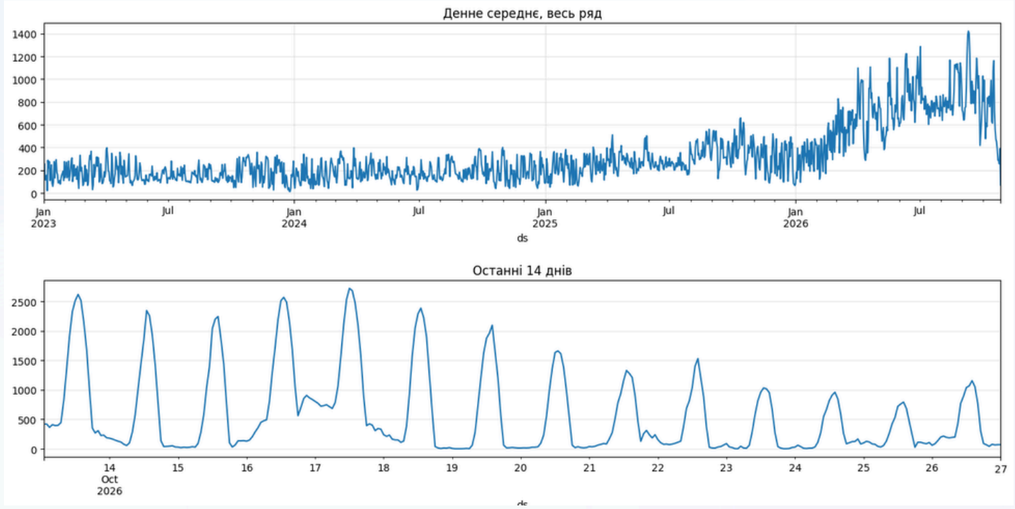
48-HOUR POWER FORECAST

FINAL PROJECT

Tetiana Mazko



THE DATA



Daily cycle

- Solar-like shape, peak at 12–14 h
- Night output is not zero (about 30–170 lately) and follows wind
- No weekday effect

Level shift

- Daily mean: 100–400 in 2023–25, 600–1400 in 2026 (probably added capacity)
- Every target is divided by the 7-day mean

Level is falling now

- Peaks fell from about 2700 (17 Oct) to 800–1150 (23–26 Oct), even on cloud-free days
- Cause unknown

33,480 hourly rows, no gaps, duplicates or negatives.

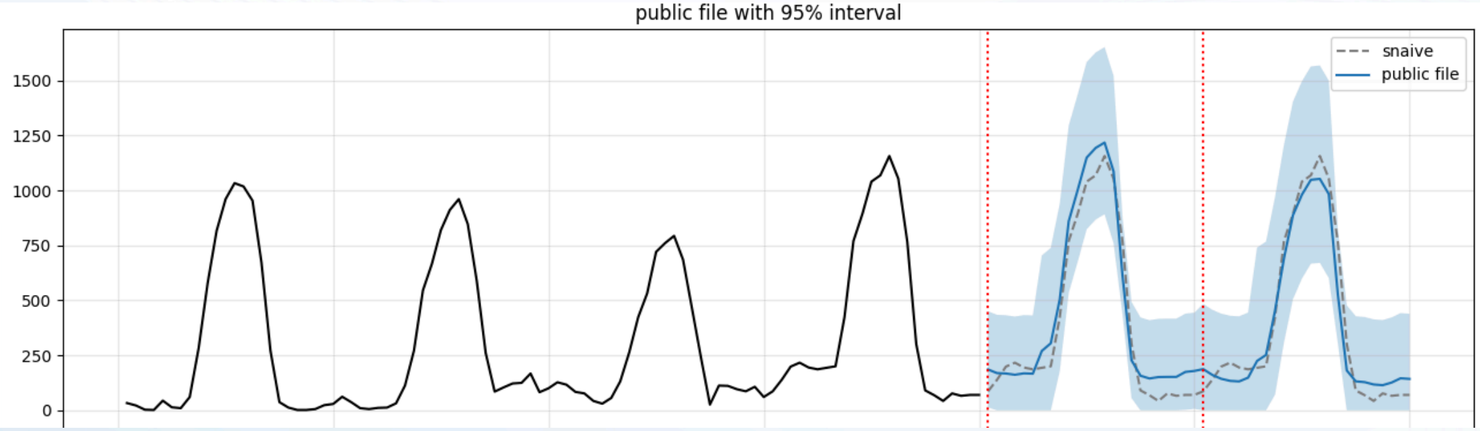
The clock change shifts the daylight window from 8–16 h to 7–15 h on day 1 of the horizon, as in earlier years.

BEFORE SUBMISSION 1

Model	Skills Score
LightGBM submited	**0.78**
LightGBM v1 (9 features)	0.885
Mean of last 3 days	1.005
LightGBM + ETS	1.147
LightGBM + SARIMAX	1.249
ETS	2.063
SARIMAX	2.121

Submitted LightGBM

- Target: next 48 h ÷ 7-day mean; one model for all hours
- 23 features: lags (same hour 1–3 days ago), recent levels and peaks, weather
- Trained on full history, recent days weighted more (half-life 180 days)
- Training set doubled with noisy weather copies (to mimic forecast error)
- Night hours × 0.8
- Interval: residual quantiles, widened ×1.6
- Public day 1: RMSE 126.9 vs 115.7 (naive), coverage 100%



PUBLIC LeaderBoard DAY1

First 24 hidden hours

#	Student	RMSE	MAPE	Winkler	Coverage	Score	Since last	vs Session 1
13	*----SNAIVE----*	*115.7*	*84.3%*	*521*	*100%*	*14.6*		
13	Tetiana Lanina	115.7	84.3%	521	100%	14.6	=	
13	Tetiana Mazko	115.7	84.3%	521	100%	14.6	=	

PUBLIC LeaderBoard DAY2

First 24 hidden hours

#	Student	RMSE	MAPE	Winkler	Coverage	Score	Since last	vs Session 1
21	Maria Nosatko	83.1	69.7%	753	100%	19.2	↓3	
22	Illia Narbut	129.1	62.2%	572	92%	19.5	↓3	+12%
23	Dmytro Tkach	189.2	48.2%	675	100%	19.9	↓1	
24	Tetiana Mazko	126.9	81.0%	549	100%	20.5	↓11	+10%

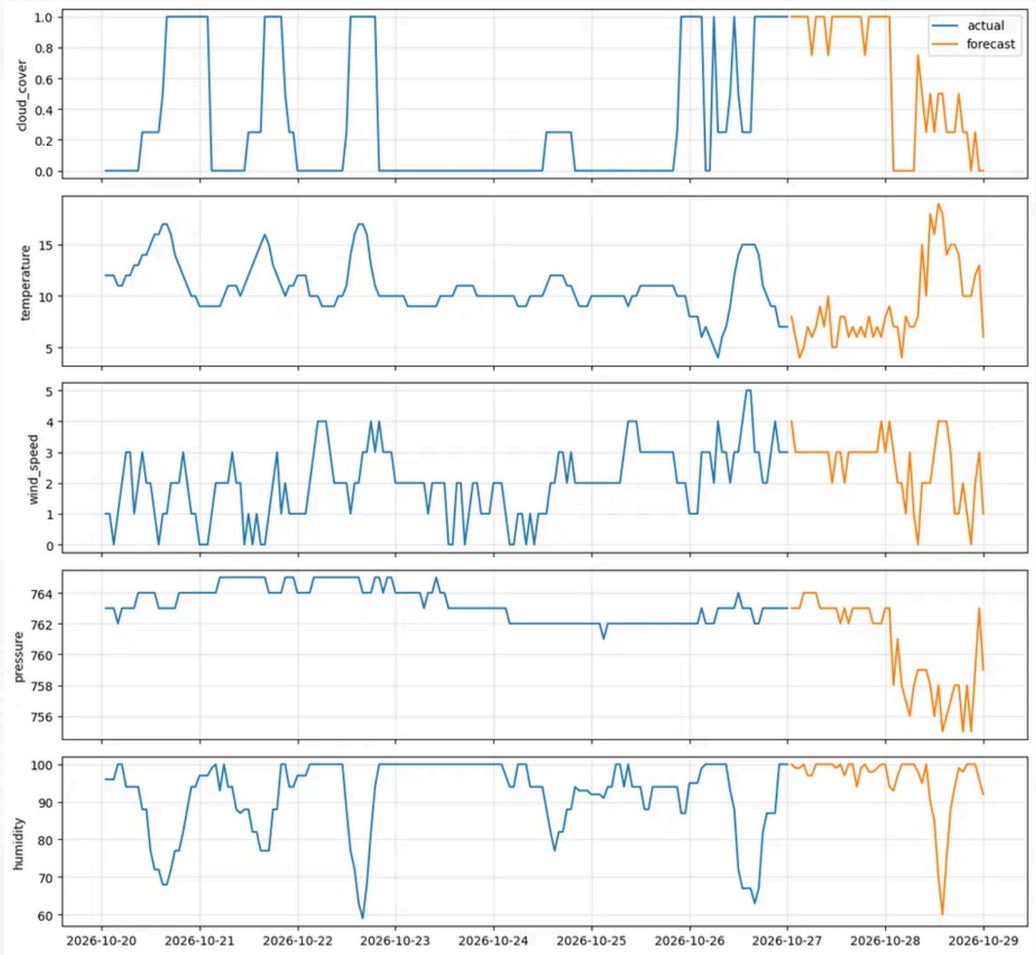
Weather EDA

- Forecast is clean
- Day 1 overcast, day 2 mostly clear; pressure drops on day 2
- Cloud does not explain daily peaks (peak ratio about 1.0 for clear, mixed and overcast days)
- Wind is the strongest weather signal: peaks (ρ = 0.46) and night output (ρ = 0.56–0.65)
- Autumn decline is typical: peaks fell to 0.53–0.69 of mid-October in 2024–26
- LightGBM reacts to cloud more than the data supports (±0.5 cloud moves the peak by 300)

What it changed

- No blanket “lower the forecast” correction
- Tested night-wind features: no gain, lags already carry it
- Tested short windows: worse
- Not tested: LightGBM without cloud

Open question: weather helps Chronos-2 a lot, though EDA shows it is secondary



LIGHTGBM: FRAMEWORK & FEATURES

Feature set	RMSE	MAPE
Seasonal naive	328.0	320.5
First feature set (A)	290.4	277.4
A + daily weather aggregates	284.2	176.5
A + extra lags and levels	270.9	246.3
All features, no decay	268.0	158.6
All features + decay	**259.3**	**140.8**

LIGHTGBM: TRAINING WINDOW

Window (low, 84)	RMSE	MAPE	Winkler
Full history	236.3	78.9	1152
365 days	**228.8**	**73.6**	**1086**
180 days	233.1	84.6	1122
120 days	245.2	93.4	1173
90 days	238.5	91.2	1191
60 days	246.2	96.7	1278
45 days	258.1	97.6	1364
30 days	258.0	101.6	1355

LIGHTGBM: OTHER EXPERIMENTS

Experiment	Result (low, day 2)	Verdict
Night multiplier 0.8 vs none	MAPE −16 [−26, −8]	0.8
Night multiplier 0.85 / 0.9 / 0.95	MAPE +3.7 / +7.6 / +11.8	rejected
Hyper-parameters (leaves, trees, λ)	RMSE 234.6–238.7 vs 236.3	noise
Sample weights (lv7, 1, lv7²)	RMSE 226.6–229.8 vs 228.8	noise
Night wind features	RMSE 229.6 / 233.0 vs 228.8	worse
Weather-noise copies in training	RMSE 235.9 vs 235.2	noise
Ensembles of LGBM windows, + mean3	RMSE 227.7–231.9 vs 228.8	noise

CHRONOS: FIRST ATTEMPTS

Model (low, day 2)	RMSE	MAPE	Winkler
Seasonal naive	292.5	120.2	1357
LightGBM (365 d)	228.8	73.6	1086
Chronos-2, no weather, C=336	271.0	98.0	1324
Chronos-2, no weather, C=1008	267.2	105.2	1271
Chronos-2, no weather, C=2400	266.6	113.4	1294
0.8 LightGBM + 0.2 Chronos-2 (no weather)	225.8	77.7	1071
Chronos-2 + weather, C=1008	**211.1**	**80.2**	**963**

C — length of context in hours

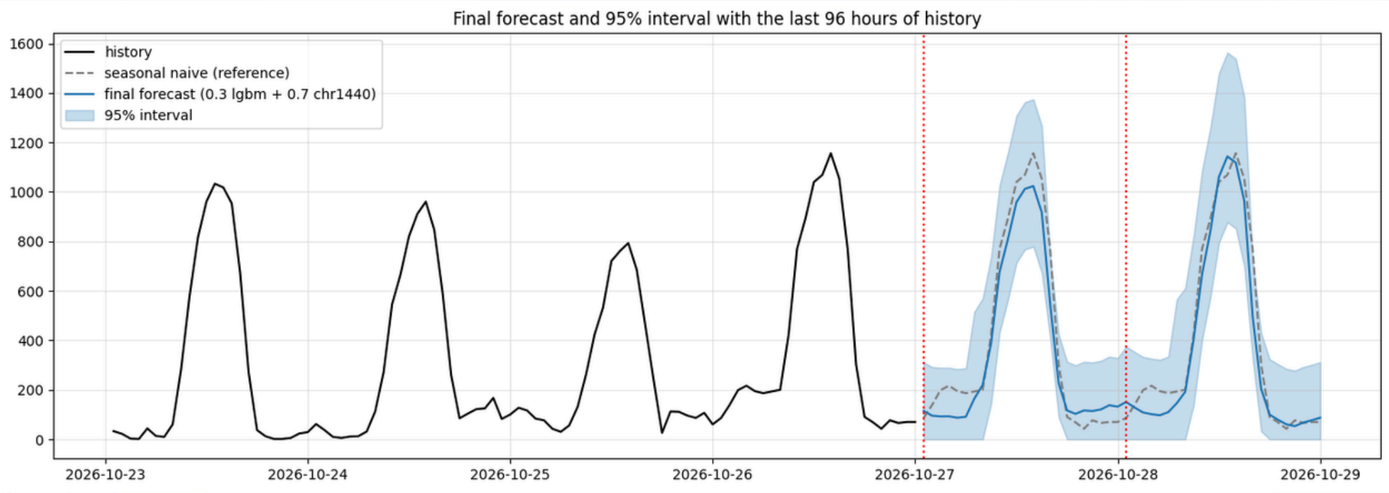
CHRONOS: CONTEXT LENGTH

	low (84)			all (240)		
Context C	**RMSE**	**MAPE**	**Wink.**	**RMSE**	**MAPE**	**Wink.**
336 h (14 d)	273.3	83.4	1314	271.8	99.2	1495
504 h (21 d)	215.9	70.7	967	227.1	78.0	1145
672 h (28 d)	214.4	76.9	965	223.8	76.4	1149
1008 h (42 d)	211.1	80.2	963	219.1	77.0	1116
1440 h (60 d)	**210.1**	**73.9**	**940**	**211.4**	**71.3**	**1087**
1800 h (75 d)	211.4	72.2	942	211.4	71.3	1082
2160 h (90 d)	210.9	74.5	961	210.0	72.2	1088
2400 h (100 d)	268.9	95.0	1288	264.3	107.9	1481

BLENDING LIGHTGBM + CHRONOS-2

	low (84)				all (240)			
LightGBM / Chronos-2	**RMSE**	**MAPE**	**Wink.**	**Skill**	**RMSE**	**MAPE**	**Wink.**	**Skill**
1.0 / 0.0	228.8	73.6	1086	0.78	231.4	90.1	1255	0.77
0.6 / 0.4	208.5	71.4	966	0.71	209.3	80.3	1126	0.69
0.5 / 0.5	205.9	71.3	947	0.70	206.6	78.3	1107	0.68
0.4 / 0.6	204.5	71.5	939	0.70	205.1	76.4	1094	0.68
0.3 / 0.7	**204.2**	**71.9**	**931**	**0.70**	**204.8**	**74.8**	**1086**	**0.68**
0.0 / 1.0	210.1	73.9	940	0.72	211.4	71.3	1087	0.70

FINAL SUBMISSION



	low (84)			all (240)		
Model (day 2)	**RMSE**	**MAPE**	**Wink.**	**RMSE**	**MAPE**	**Wink.**
Seasonal naive	292.5	120.2	1357	302.3	194.1	1543
Final blend	**204.2**	**71.9**	**931**	**204.8**	**74.8**	**1086**

Skill (RMSE): 0.70 (low) and 0.68 (all); Winkler 31% lower (low)

- 0.3 × LightGBM (365 days, 3 seeds) + 0.7 × Chronos-2 (weather, C=1440 h)
- Night hours × 0.8
- Interval
  - Quantiles of out-of-fold residuals of the blend, by day and daylight
  - Widened ×1.2, lower bound 0
  - Validation coverage: 0.94 (low), 0.97 (all)

LIMITS

What did not work

- ETS and SARIMAX (RMSE about 2× the benchmark)
- Chronos without weather
- Short LightGBM windows
- Tuning

Main risks

- Size of weather-forecast error is guessed
- Settings chosen on the same 240 origins
- Current level is below what the model saw in validation

What remains

- Public day 1 is one day, weak evidence
- Chronos-2 is a pretrained model, I cannot inspect its logic
- Inference is computationally heavier

QUESTIONS

Thank you for your attention!

git hub: https://github.com/tmazko/Time_Series_Final

